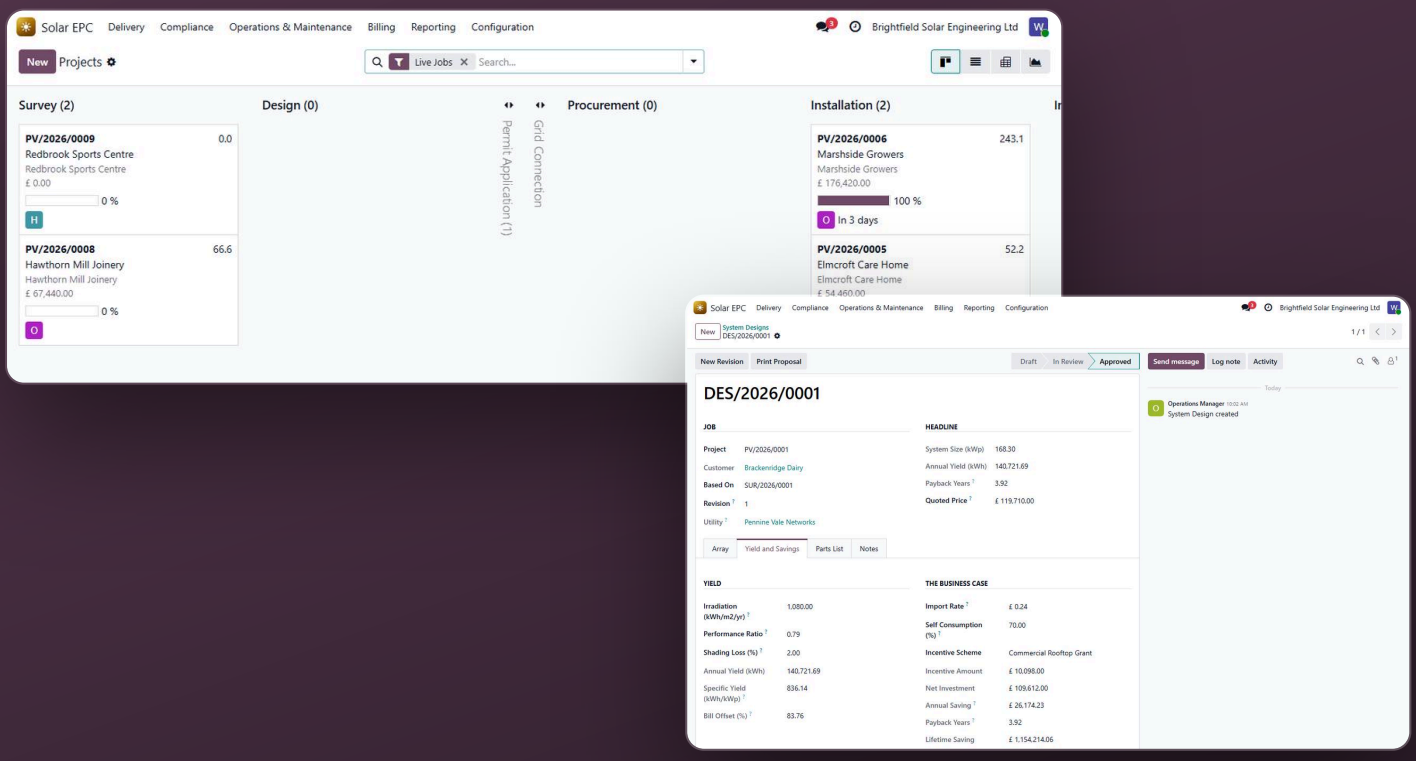
WAY4TECH SOLUTIONS

Live demo
solar-demo.way4tech.com
w4demo / Demo2026 · Odoo 19, Community

COMPLETE GUIDE · THE TRADE, SETUP AND EVERY SCREEN

Solar Project Management for Odoo 19

Solar EPC from the site survey to the meter: string sizing, yield and payback, the parts list, milestone billing, grid applications, commissioning, warranties, O&M and the generation log. Every screen and every step, shown on a live demo. No Odoo background needed.



START

About this guide

A solar job is sold on a spreadsheet and built on a roof, and the two drift apart unless one record carries the job from the first survey to the last meter reading. This module is that record. The guide walks it screen by screen, on a live demo. You do not need to have used Odoo before.

Who it is for

The owner of a solar installer or EPC contractor, the design engineer who sizes and prices the system, the project manager who runs the job, the crew supervisor, and the O&M team that looks after the sites afterwards.

How to read it

Front to back the first time. Each part opens with what you will learn and a short thing to try on the live demo, then walks every field and every button.

The live demo

Brightfield Solar Engineering Ltd, an invented installer in Preston with two crews, ten jobs at every stage from scoping to handed over, and up to ten months of meter readings on the finished sites.

HOW IT WORKS

The demo installer

Address	**solar-demo.way4tech.com**
Sign in	**w4demo** / **Demo2026**
Installer	Brightfield Solar Engineering Ltd · Preston · pounds sterling
Grid and grants	Pennine Vale Networks (an invented utility) · a Commercial Rooftop Grant from an invented energy agency
Runs on	Odoo 19, Community edition, with Inventory and Invoicing

Every company, utility, person and number on the demo is invented. The demo is shared and resets from time to time, so anything you change may be changed back or seen by the next person.

What is in the guide

PART 1 How a solar EPC job runs

PART 2 The business at a glance

PART 3 Sign in, then set up from zero

PART 4 The project and its stages

PART 5 The site survey

PART 6 The system design: strings, yield, price and payback

PART 7 The contract: proposal, payment schedule and documents

PART 8 On site: materials and the daily site diary

PART 9 Grid connection and net metering

PART 10 Commissioning

PART 11 Handover and the warranty register

PART 12 Operations and maintenance contracts

PART 13 The generation log and underperforming sites

PART 14 Costs, margin and reporting

PART 15 Printed documents, emails and the nightly jobs

PART 16 Who sees what, a walk-through and questions

PART 17 The menu map

PART 1 How a solar EPC job runs

Five minutes on the trade, so the screens make sense.

IN THIS PART YOU WILL LEARN

- Survey, design, permit, grid, build, commission, hand over
- Why the design is where the sale is made
- Why the grid connection is the longest pole
- What has to be proved before a system is energised
- What happens for the next twenty five years

From roof to meter

An EPC contractor (engineering, procurement and construction) takes a solar system from an empty roof to a working, certified, metered installation. Somebody **surveys** the site: the area, what is in the way, which way it faces, how steep it is, what shades it, whether the structure will take the frame, and a year of electricity bills. A **design** turns that into a module count, an inverter, a string layout, a yield and a price. The customer signs. Then come the paperwork stages: a building or roof **permit** where one is needed, and the **grid connection** or net metering application to the utility, which on most jobs takes longer than everything else put together.

Build, prove, hand over

Materials are bought and called off to site, the crew installs, somebody inspects, and before the system is energised every string is **commissioned**: open circuit voltage against what the design said, insulation resistance, polarity, earthing. An electrical safety certificate is issued. Only then is the system **handed over** with its warranties: one for each kind of component, because modules, inverters and the workmanship are covered for different terms by different people.

Afterwards

Many installers sell an **operations and maintenance** contract: scheduled visits, cleaning, string tests, a response time and a performance guarantee. The meter tells you every month whether the site is making what the design said it would.

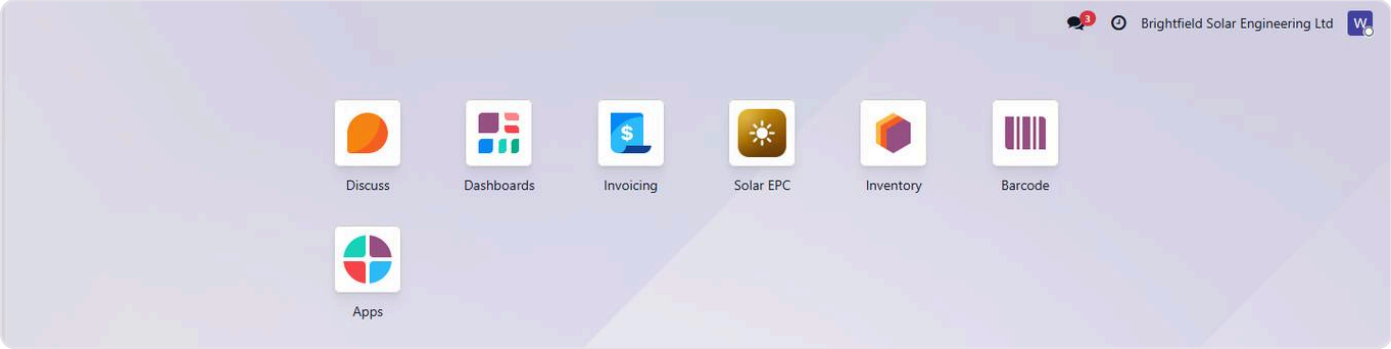
One job



WORD	MEANING
kWp	Kilowatt peak: the size of the array, module count times the watts of each module
Voc	Open circuit voltage. It rises as the cells get colder, which is why strings are sized at the coldest morning
MPPT	An input on the inverter that tracks the best operating point for the strings on it
Performance ratio (PR)	What the system actually delivers against what the sunshine could give it
Specific yield	kWh a year per kWp installed
Net metering	An arrangement where exported energy is credited against imported energy
Stage gate	The rule that a job cannot leave a stage until that stage's document is approved

PART 2 The business at a glance

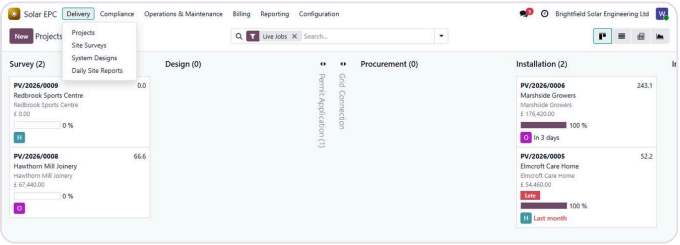
The module adds one app, Solar EPC, with seven menus, and three roles that decide who sees what.



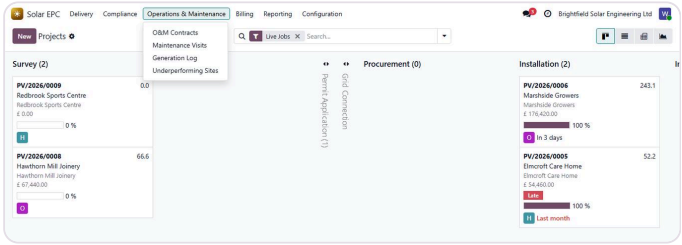
The Solar EPC app on the Odoo home screen.

The seven menus

MENU	WHAT IS UNDER IT
Delivery	Projects, Site Surveys, System Designs, Daily Site Reports
Compliance	Grid Applications, Project Documents, Commissioning, Warranty Register
Operations & Maintenance	O&M Contracts, Maintenance Visits, Generation Log, Underperforming Sites
Billing	Payment Schedule, Project Costs
Reporting	Margin by Project, Maintenance Revenue
Configuration	Settings, Project Stages, Utilities, Incentive Schemes, Installation Crews, Solar Components



Delivery dropped down over the project board.



Operations & Maintenance.

Three roles

Site Technician

Sees the jobs, site reports, visits and commissioning tests of their own crew. Builds to the design, but sees no prices, costs or contract values.

Design Engineer

Every job. Surveys, designs and prices, the payment schedule, O&M contracts and underperforming sites.

Solar Manager

Everything: margins, project costs, reporting and configuration. The administrator gets this role on install.

> The demo login w4demo has the Solar Manager role. It is not an administrator and cannot open Settings.

PART 3 Sign in, then set up from zero

Sign in to the demo, then see what a new installer sets up before the first survey. On the demo all of this is done.

TRY IT YOURSELF ON THE DEMO

1. Open solar-demo.way4tech.com and sign in with w4demo and Demo2026
2. Open Configuration, Project Stages and open Installation
3. Open Configuration, Solar Components and a module's Solar tab
4. Open Configuration, Utilities, Pennine Vale Networks



1

Go to the demo

Open **solar-demo.way4tech.com** and sign in as `w4demo` with password `Demo2026`.



2

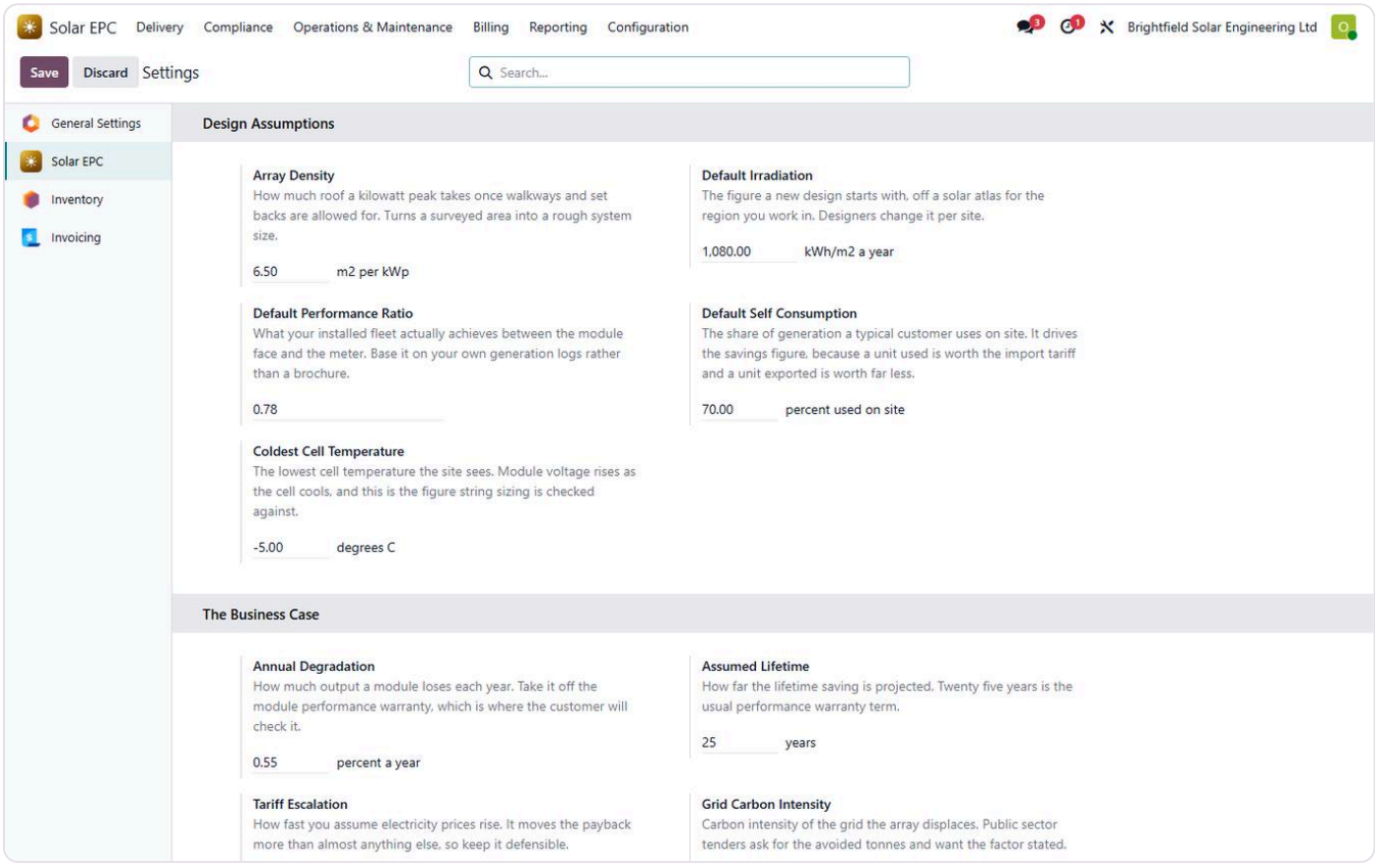
Find Configuration

Open **Solar EPC**, then Configuration.

Install the app

The module is **Solar Project Management** (technical name wt_solar_epc, USD 349 on the Odoo App Store, version 19.0.1.0.0). It brings in Inventory and Invoicing, both part of Odoo Community: the parts list is called off the warehouse and milestones are invoiced. It installs the nine project stages, five scheduled jobs, three email templates and four printed documents.

The settings

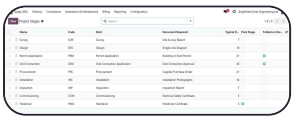


Settings, Solar EPC, opened with the administrator, because the demo login cannot reach Settings.

SETTING	WHAT IT DOES	DEFAULT
Array Density (m2 per kWp)	How a survey turns usable area into a rough system size	6.5
Default Irradiation	kWh/m2 a year a new design starts with	1,080
Default Performance Ratio	The PR a new design starts with	0.78
Default Self Consumption (%)	Share of the yield used on site rather than exported	70
Annual Degradation (%)	Yield lost each year in the payback walk	0.55
Assumed Lifetime (years)	Years the payback and lifetime saving run over	25
Tariff Escalation (%/yr)	How fast the import tariff rises in the payback walk	5
Grid CO2 (kg/kWh)	For the carbon avoided figure	0.45
Coldest Cell Temperature (C)	The morning strings are sized for	-5
Structural Note Required	A survey cannot be signed off without one	on
Enforce the Stage Gate	A job cannot leave a stage without its approved document	on
Grid Approval Before Handover	Hand over refused without an approved connection	on
Minimum Insulation (Mohm)	Below this a string fails commissioning	1.0
Voc Tolerance (%)	Measured Voc must be within this of the calculated	10
Workmanship Warranty (years)	The installer's own cover at handover	5
Warranty / Document / Visit warnings (days)	How early the jobs warn	60 / 30 / 7
Underperformance Threshold (%)	How far below estimate a month can fall before it is flagged	10
Milestone Product	The service product milestone invoices are raised on	set by the demo

Project stages

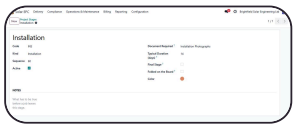
Nine stages are installed: Survey, Design, Permit Application, Grid Connection, Procurement, Installation, Inspection, Commissioning and Handover. Each names the **one document** it will not let a job leave without (a survey report, a single line diagram, a permit, the grid approval, a supplier purchase order, installation photographs, an inspection report, the electrical safety certificate, the handover certificate) and a target number of days. Only one stage may be the closing stage.



3

Configuration, Project Stages

Code, kind, the required document and the target days for each.



4

One stage

Installation needs Installation Photographs. The target days of the remaining stages add up to the target handover date when a contract is signed.

Solar components

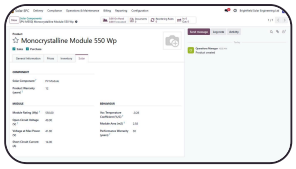
Modules, inverters and the balance of system are ordinary Odoo products with a **Solar** tab. The component type decides what the design can pick it for.



5

Configuration, Solar Components

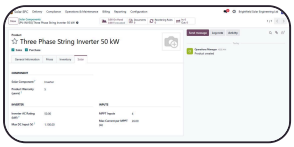
The demo carries two modules, two inverters, a mounting set, DC cable, a protection set, an export meter and the milestone service product.



6

A module

Watt peak, Voc, Vmp, Isc, the temperature coefficient of Voc and the area, with the product and performance warranty years. A module without a Voc, or a temperature coefficient that is positive, is refused.

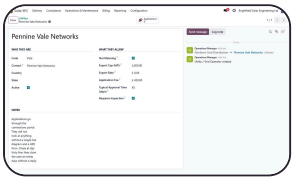


7

An inverter

AC kW, the maximum DC voltage, the number of MPPT inputs and the maximum current per input.

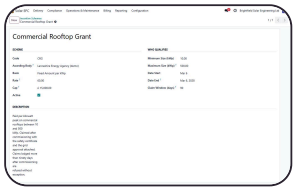
Utilities, incentives and crews



8

A utility

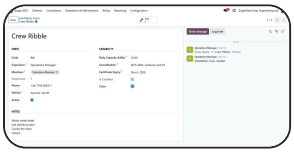
Whether it offers net metering, the largest system it accepts, the export tariff, the application fee, how many days it usually takes to approve, and whether it inspects before approving. The export cap refuses an application above it and the lead days set the follow up date.



9

An incentive scheme

Paid per kWp, as a percentage of the contract, or as a fixed amount, with a cap, a kWp range, the dates it runs and the days allowed to claim. A design outside the range or the dates simply shows no incentive.



10

A crew

Supervisor and members, vehicle, how many kWp it puts up in a day (which gives the crew days on each job) and its certification with an expiry. A crew whose accreditation has lapsed cannot be put on a job.

PART 4 The project and its stages

Every job is one project record. It moves through two things at once: the commercial state (scoping, quoted, signed, on site, commissioned, handed over) and the nine delivery stages, each with its document gate.

IN THIS PART YOU WILL LEARN

- The project board and the list
- Commercial states against delivery stages
- The stage gate and why it cannot be dragged past
- What the project form shows on each tab

TRY IT YOURSELF ON THE DEMO

1. Open Delivery, Projects: the board opens on Live Jobs
2. Open PV/2026/0001, Brackenridge Dairy, handed over
3. Open PV/2026/0005, Elmcroft Care Home, on site and late
4. Try dragging a Survey card straight to Installation (it is refused)



1

Delivery, Projects

The board has one column per stage, opened on Live Jobs. Each card shows the job, the customer, the size in kWp, the contract value, installation progress and the activity due. A red Late tag means today is past the target handover.



2

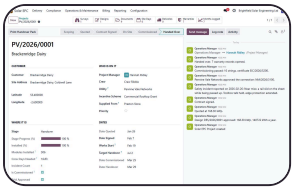
The list

Customer, site, stage, state, kWp, contract value and dates, for filtering and grouping.

STATE	REACHED BY	MEANS
Scoping	creating the job	survey and design under way
Quoted	Send Proposal	an approved design has been sent
Contract Signed	Contract Signed	payment schedule locked, target date set
On Site	Start on Site	a crew is building
Commissioned	Commissioned	a commissioning test has passed
Handed Over	Hand Over	warranties opened, the job is closed
Cancelled	Cancel	refused once any milestone is invoiced

The stage gate

Next Stage moves a job on one stage. With **Enforce the Stage Gate** on in Settings, a job cannot leave a stage until that stage's document has been filed and approved on its Documents tab. The check runs however the stage changes, so dragging a card on the board is refused the same way, and every stage jumped over is checked too: dragging from Survey to Handover asks for every document in between. Moving a job backwards is always allowed, because rework happens.

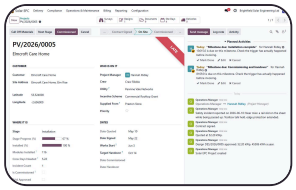


3

A handed over job

PV/2026/0001: customer, site, coordinates, project manager, crew, utility, incentive scheme and the store it is supplied from; the approved design with its kWp and yield; contract value, estimated and actual cost, margin, invoiced and still due; dates quoted, signed, works start, target, commissioned and handed over.

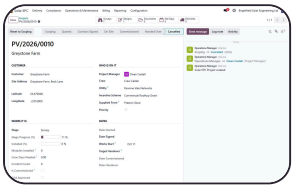
Stat buttons open the surveys, designs, documents, site reports, material movements, warranties and generation of the job.



4

A job on site

Elmcroft Care Home: on site, installation complete according to the site diary, but past its target handover, so it is Late.



5

A cancelled job

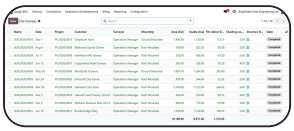
Greystone Farm was cancelled before signing. **Reset to Scoping** reopens a cancelled job. A job with milestone invoices cannot be cancelled until they are credited, and a handed over job cannot be cancelled at all.

PART 5 The site survey

The survey is the evidence the design rests on: how much roof, what is in the way, which way it faces, whether it will hold, and what the customer uses.

TRY IT YOURSELF ON THE DEMO

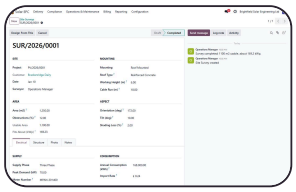
1. Open Delivery, Site Surveys
2. Open SUR/2026/0001 and look at the Electrical and Structure tabs



1

Delivery, Site Surveys

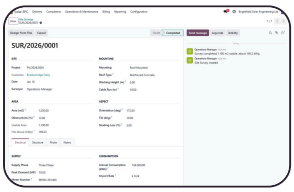
Every survey with its area, the size it fits, consumption and state.



2

Step 1: the roof

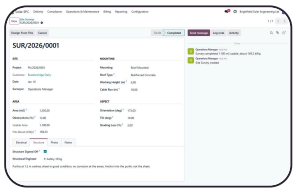
Pick the project and the surveyor. **Mounting** is roof, ground, carport or facade; **Roof Type** one of reinforced concrete, metal sheet, tiled pitch, flat membrane, asbestos sheet, ground screw or ballasted. Enter the **Area** and the **Obstructions** percentage: the usable area and the rough size it fits (usable area over the area per kWp in Settings) are worked out. Orientation (0 to 360 degrees), tilt (0 to 90), shading loss and working height complete the roof.



3

Step 2: the bills

Annual Consumption, peak demand, the **Import Rate**, single or three phase supply, the meter number and the cable run to the panel. Without the consumption and the tariff there is no honest savings figure.



4

Step 3: the structure

Tick **Structure Signed Off** with the engineer's name and a structural note. With Structural Note Required on, **Sign Off** is refused without the note: nobody puts a frame on a roof that was never looked at.

Design From This opens a new design pre-filled from the survey: shading, import tariff and cable run.

A signed off survey cannot be cancelled while an approved design is built on it. A cancelled survey is not reopened: copy it and survey again.

PART 6

The system design: strings, yield, price and payback

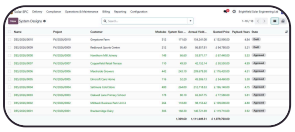
The design is where the sale is made. It carries the module and the inverter as real products, sizes the strings against the inverter's voltage window, works out the yield, builds the parts list, prices the job and walks the payback year by year.

IN THIS PART YOU WILL LEARN

- Module count, inverter count and the DC to AC ratio
- Size Strings and the cold morning check
- The yield and the bill offset
- Rebuild Parts List, cost, price and price per watt
- Payback, lifetime saving and carbon
- Review, approval and revisions

TRY IT YOURSELF ON THE DEMO

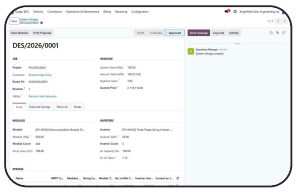
1. Open Delivery, System Designs, then DES/2026/0001
2. Open its Array tab and look at the voltage per string
3. Open Yield and Savings and Parts List
4. Open the draft design for Redbrook Sports Centre



1

Delivery, System Designs

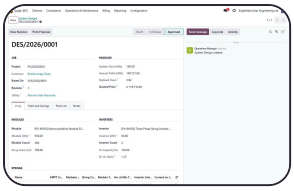
Every design and revision, with kWp, yield, price, payback and state.



2

Step 1: module and inverter

Pick the project and its signed off survey, the module and how many, the inverter and how many. The design works out the **System Size** in kWp, the AC capacity, the **DC/AC ratio**, the array area and whether it fits the usable area from the survey. Approval is refused if it does not fit.

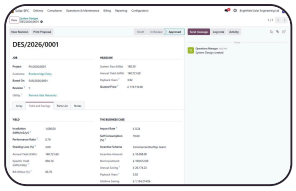


3

Step 2: Size Strings

Press **Size Strings**. The module's Voc is raised to the coldest cell temperature in Settings (Voc rises as cells cool), the longest string that stays inside the inverter's maximum DC voltage is chosen, and the modules are spread evenly over the fewest strings, assigned across the MPPT inputs. Each string row shows its cold Voc, Vmp and the current on its input, and flags over voltage or over current.

The strings must add up to the module count, or the design is refused. A string can be edited by hand.

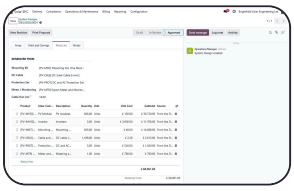


4

Step 3: yield and savings

Annual Yield = kWp × irradiation × performance ratio × (1 - shading loss). The specific yield and the share of the customer's consumption it covers follow. The business case walks every year of the lifetime: yield falls by the degradation rate, the tariff rises by the escalation rate, the self consumed share is valued at the import rate and the exported share at the utility's export tariff where it offers net metering. That gives the first year saving, the **payback** in years (counted from the net investment after any incentive), the lifetime saving and yield, and the CO2 avoided.

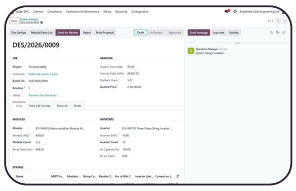
There is no weather model behind the yield: irradiation from an atlas, a PR from experience and the measured shading, stated plainly enough for a customer to check.



5

Step 4: Rebuild Parts List

Press **Rebuild Parts List**. It writes the modules, the inverters, one mounting set per module, DC cable for the run there and back on every string plus the jumpers, one protection set per inverter and one meter, each at the product's cost. Lines you add by hand survive a rebuild. Labour cost and overhead percentage complete the **estimated cost**; enter the **Quoted Price** and the margin and the price per watt peak are shown.



6

Step 5: review and approval

Send for Review needs a parts list and a price. **Approve** makes this the design the job is built to: any earlier approved design is superseded, and the project takes the design and its price as the contract value. **Reject** sends it back. **New Revision** copies an approved design as the next revision; an approved design cannot be edited or reopened.

Print Proposal prints the customer proposal (Part 15).

Costs, prices and margins on the design are hidden from Site Technicians: they see the array, the strings and the parts list, which is what they build from.

PART 7 The contract: proposal, payment schedule and documents

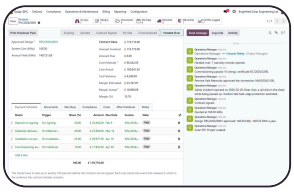
An approved design is quoted, the customer signs against a payment schedule that has to add up to exactly 100 percent, and from then on each stage asks for its document.

TRY IT YOURSELF ON THE DEMO

1. Open PV/2026/0001 and its Payment Schedule tab
2. Open Billing, Payment Schedule
3. Open Compliance, Project Documents

Send Proposal and sign

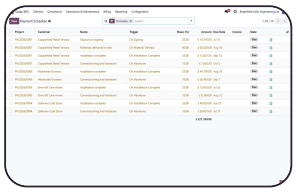
Send Proposal on a scoping job with an approved design moves it to Quoted, stamps the date and queues the Proposal Sent email if the customer has an address. **Contract Signed** needs an approved design, a contract value, and a payment schedule whose shares add up to exactly 100 percent. It stamps the signing date and sets the **target handover** from the target days of the stages still ahead.



1

The payment schedule

One line per milestone with a **trigger** (on signing, design approval, permit granted, material delivery, installation complete, commissioning, handover), a share, the amount (share of the contract value), a due date and a state: Pending, Due, Invoiced, Paid or Cancelled. Raise the invoice from the line.



2

Billing, Payment Schedule

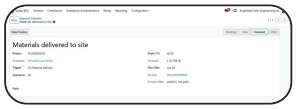
Every milestone across the jobs.



3

A milestone due

The nightly milestones job moves pending milestones whose due date has come to Due on signed, on site and commissioned jobs, and leaves the project manager a to do to check the trigger really happened. **Mark Due** does it by hand.



4

Invoice

Invoice raises a customer invoice for the milestone amount on the milestone product in Settings, with its taxes put through the customer's fiscal position, in the company currency. A job still scoping or quoted cannot be invoiced.

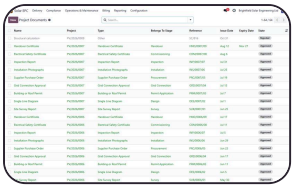


5

Paid

The same nightly job marks a milestone Paid once its invoice is posted and paid in Accounting.

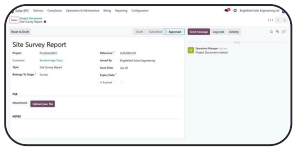
Project documents



6

Compliance, Project Documents

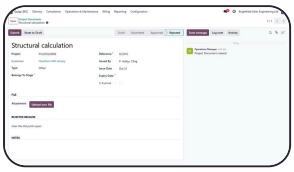
Seventeen document types, from site survey report and single line diagram to safety certificate, handover certificate, warranty pack and incentive claim.



7

A document

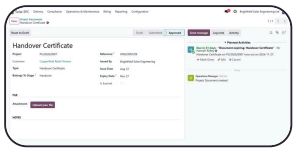
The stage it belongs to, the reference as the issuer wrote it, the issuer, issue and expiry dates, an attachment and a state: **Submit**, then **Approve**. An expired document cannot be approved.



8

Rejected

Write the reason, then **Reject**. Reset to Draft and submit again once it is fixed.



9

About to lapse

The weekly documents job leaves a to do on any approved document expiring within the warning days (30), on jobs still open.

PART 8 On site: materials and the daily site diary

Once the contract is signed and the crew is booked, the job lives on the site diary: who was there, what went up, what stopped them and whether anybody got hurt.

TRY IT YOURSELF ON THE DEMO

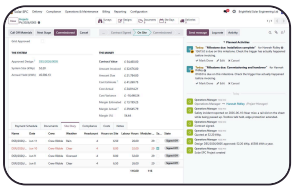
1. Open PV/2026/0005 and its Site Diary tab
2. Open Delivery, Daily Site Reports and the draft one for today

Start on Site needs a signed contract and a crew. It moves the job to On Site and stamps the works start if it was empty. The crew days needed are the system size over what the crew puts up in a day.

Calling materials off

Call Off Materials is meant to create a delivery from the job's warehouse to the customer with every stocked line of the parts list, linked to the job, so the store knows which roof the pallet is for. The Materials stat button lists those movements.

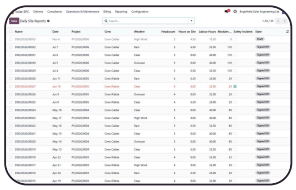
> In this version Call Off Materials stops with an error on Odoo 19 (it writes a field the stock move no longer has), so the delivery has to be created in Inventory by hand for now. We have reported it for a fix.



1

The Site Diary tab

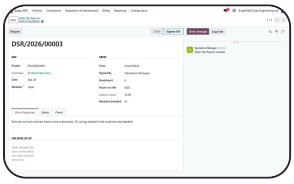
Every report on the job, with modules installed, hours and any incident. The job rolls them up into modules installed and installation progress against the design's module count.



2

Delivery, Daily Site Reports

All crews, all days.

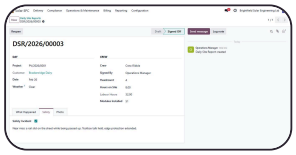


3

A report

Date, crew, who signed it, weather (clear, overcast, rain, high wind, extreme heat, frost), headcount, hours on site (giving labour hours), modules installed, what was done and what is holding the crew up, so the office can clear it before the next day.

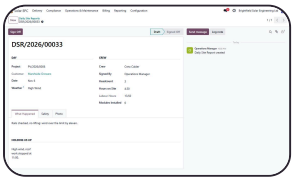
The modules installed on all signed off reports cannot add up to more than the design's module count.



4

Safety

Tick Safety Incident and describe it. Incidents are counted on the job.



5

Sign Off

A report counts once it is **Signed Off**. Reopen puts it back to draft. Print gives the daily site report (Part 15).

PART 9 Grid connection and net metering

The connection is the longest pole on most jobs. Each application records what was applied for, when the utility should have answered, the inspection and the approval number.

TRY IT YOURSELF ON THE DEMO

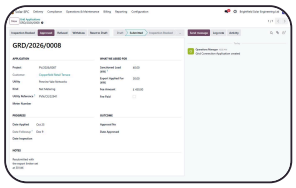
1. Open Compliance, Grid Applications
2. Open the refused one for Copperfield Retail Terrace, then the resubmission



1

Compliance, Grid Applications

Opens on With the Utility: applications lodged and not yet answered, with the follow up date. Two here are past it, which is the chase list.

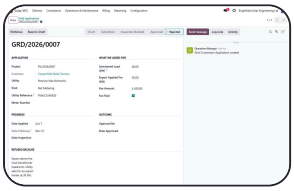


2

Step 1: Lodge

Pick the utility and the kind (grid connection, net metering or capacity upgrade), the utility reference, the meter number, the sanctioned load and the **Export Applied For**, and the fee. **Lodge** needs an export figure. The **follow up** date is the date applied plus the utility's usual approval days.

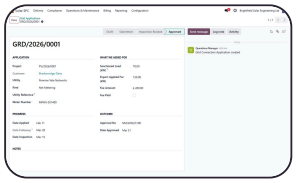
An application above the utility's maximum system size is refused.



3

Refused

Write the utility's reason and press **Refused**. Here the export was above the transformer headroom, and the job was resubmitted at 30 kW with an export limiter. **Withdraw** takes an application back.



4

Step 2: inspection and approval

Inspection Booked records the inspection date. **Approved** needs the approval number, and where the utility inspects, the inspection first. An approved application ticks Grid Approved on the job.

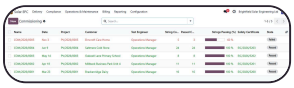
With Grid Approval Before Handover on in Settings, a job with a utility cannot be handed over until it has an approved application.

PART 10 Commissioning

Nothing is handed over until every string has been measured and passed. The test sheet is laid out from the approved design, so the tester has an expected voltage for each string before they climb up.

TRY IT YOURSELF ON THE DEMO

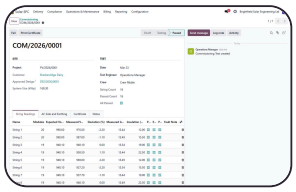
1. Open Compliance, Commissioning, COM/2026/0001 (passed)
2. Open the failed test on Elmcroft Care Home



1

Compliance, Commissioning

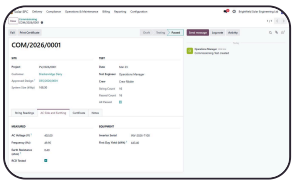
Every test with its strings, result and certificate.



2

Step 1: Open Test Sheet

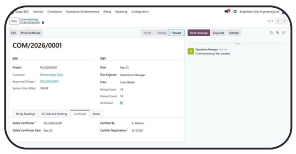
Pick the job, date and crew and press **Open Test Sheet**: one row per string on the approved design, each with the calculated Voc. A design with no strings cannot be tested. On each row enter the **measured Voc**, Isc, **insulation resistance**, polarity and earth continuity. A row passes when the measured Voc is within the tolerance (10 percent) of the expected figure, the insulation is at or above the floor (1 Mohm), both from Settings, and polarity and continuity are confirmed. A failing row says why.



3

Step 2: the AC side

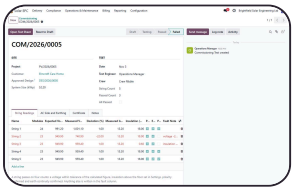
AC voltage, frequency, earth resistance, whether the RCD was tested, the inverter serial, and the yield on the day of energising.



4

Step 3: the certificate

The electrical safety certificate number and date, and who certified it with their licence number. **Pass** is refused while any string has not passed, without the certificate number, or with the AC side unmeasured.



5

Fail

Write what failed, so the crew going back knows where to look, and press **Fail**. Here one string read 22 percent low and another showed 0.6 Mohm. Open the test sheet again after the fix.

Once a test has passed, **Commissioned** on the job moves it to Commissioned. A passed test cannot be reset; on a handed over job it cannot be failed either: raise a warranty claim instead. Print Certificate prints the commissioning certificate (Part 15).

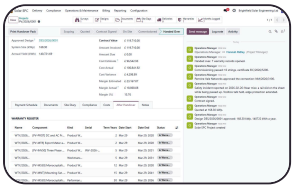
PART 11 Handover and the warranty register

Hand Over closes the job and opens one warranty for each kind of component, because a module, an inverter and the installation itself run for different terms and are honoured by different people.

TRY IT YOURSELF ON THE DEMO

1. Open PV/2026/0001, After Handover tab
2. Open Compliance, Warranty Register and the inverter with a claim

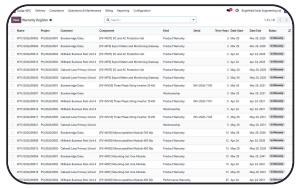
Hand Over needs a commissioned job, a passed test with a safety certificate number, and (with the setting on) an approved grid application. It moves the job to Handed Over at the closing stage and registers the warranties: a product warranty for every line of the parts list whose product carries warranty years, a performance warranty for the modules, and the installer's own workmanship cover (5 years by default). The inverter warranty carries the serial from commissioning. A handed over job cannot be moved on with Next Stage or cancelled.



1

After Handover

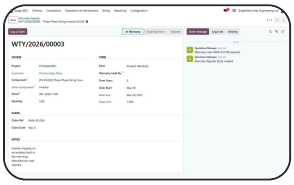
The job's warranties, O&M contract and generation log.



2

Compliance, Warranty Register

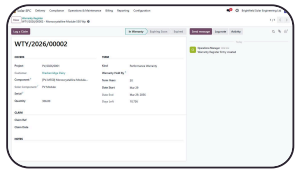
Every warranty with its component, term, end date, days left and status.



3

Log a Claim

Enter the manufacturer's case reference and press **Log a Claim**. An inverter claim needs the serial number; an expired warranty cannot be claimed against.



4

A performance warranty

The module manufacturer's output guarantee, usually far longer than the product warranty. A term above 40 years is refused as a typing mistake.

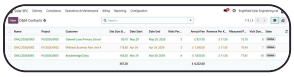
The weekly **warranty cover running out** job refreshes the status of every warranty on a handed over job and leaves a to do on any expiring within the warning days (60) that has no claim. Print Handover Pack prints the handover document (Part 15).

PART 12 Operations and maintenance contracts

An O&M contract lays out its own visit calendar, carries a response time and a performance guarantee, and tells you what the book is worth per kilowatt under contract.

TRY IT YOURSELF ON THE DEMO

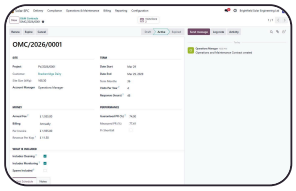
1. Open Operations & Maintenance, O&M Contracts, OMC/2026/0001
2. Open its Visit Schedule tab
3. Open Maintenance Visits and the missed one



1

O&M Contracts

Engineers and managers. Each contract with its site, kWp, fee, billing, guaranteed PR and visits.

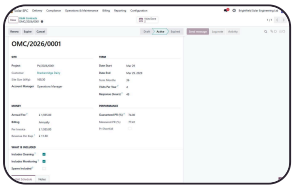


2

A contract

The handed over job (a contract can only be opened on one), the start and end dates, **visits per year**, the **fee** and billing (monthly, quarterly or annually, giving the fee per period), the **revenue per kWp**, the response time in hours, the **guaranteed PR** against the PR actually measured from the generation log, and whether cleaning, monitoring and spares are included. Two contracts on one site cannot overlap.

A PR below the guarantee is flagged as a shortfall.

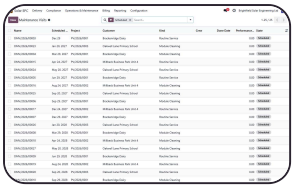


3

Activate

Activate needs a fee. It spreads the visits evenly over the term; with cleaning included, every other visit is a cleaning visit, because those need different kit on the van. **Renew** opens the next term the day after this one ends and expires the current one. **Expire** cancels the visits still scheduled. **Cancel** is refused once a visit has been done.

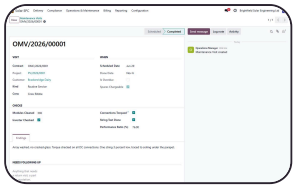
Visits



4

Maintenance Visits

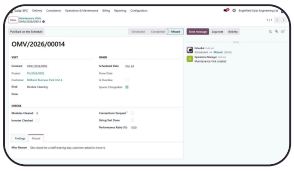
Every visit by date and state.



5

Completed

Modules cleaned, inverter checked, connections torqued, string test done, PR measured, the **findings** and any action required. **Completed** is refused without findings (a visit with no findings cannot be told from one that never happened), and a cleaning visit needs the modules washed. An action required becomes a to do on the job.



6

Missed

Write why and press **Missed**. **Put Back on the Schedule** returns a missed or cancelled visit.

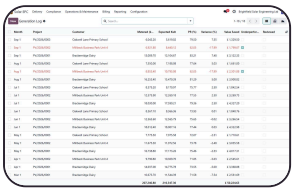
The **maintenance visits coming up** job (daily, ships off) warns the account manager about visits inside the reminder days (7) and ones gone by, and can email the customer with the Maintenance Visit Due template.

PART 13 The generation log and underperforming sites

One line per site per month: what the meter says against what the design said. A site that falls further behind than the threshold is flagged.

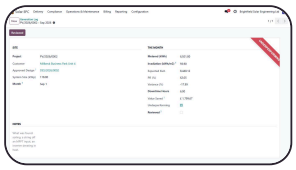
TRY IT YOURSELF ON THE DEMO

1. Open Operations & Maintenance, Generation Log
2. Open Underperforming Sites



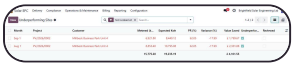
1 Generation Log

Month, metered kWh, irradiation, expected kWh, PR, variance and the value saved at the import rate. One line per site and month.



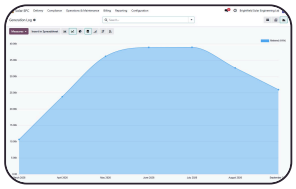
2 A month below estimate

Where the month's irradiation is entered, expected is kWp times that irradiation times the design PR and shading, and the PR is a proper measured ratio. Without it, the annual estimate is pro rated by the days in the month, which flatters summer and punishes winter. A variance below minus the threshold in Settings (10 percent) is **Underperforming**. Downtime hours and a note explain it; **Reviewed** marks it looked at.



3 Underperforming Sites

Engineers and managers: every flagged month not yet reviewed. On the demo, Millbeck Business Park dropped to 82 percent of estimate for two months.



4 The graph

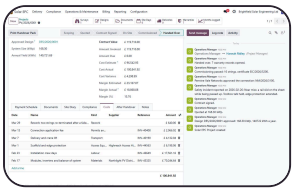
Metered against expected over time.

The **sites generating below estimate** job (weekly, ships off) raises the flagged sites with the project manager and can email the customer with the Site Underperforming template.

> Meter readings are entered or imported by hand. The module does not connect to inverter or monitoring portals.

PART 14 Costs, margin and reporting

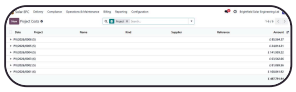
Margin per job against the estimate, with costs booked as they are spent, and maintenance revenue against the sites under contract.



1

Costs on the job

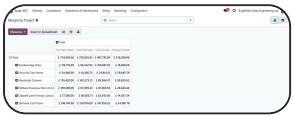
Each cost line has a kind (materials, labour, subcontractor, access equipment, transport, permits and fees, rework, other), an amount, the supplier, product and the supplier's reference. A negative cost is refused; book a credit as its own line.



2

Billing, Project Costs

Managers only: every cost across the jobs.



3

Reporting, Margin by Project

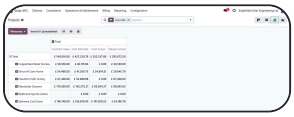
Contract value, estimated cost from the approved design, actual cost, the variance and the margin by job.



4

Reporting, Maintenance Revenue

Contract fees and kWp under contract.



5

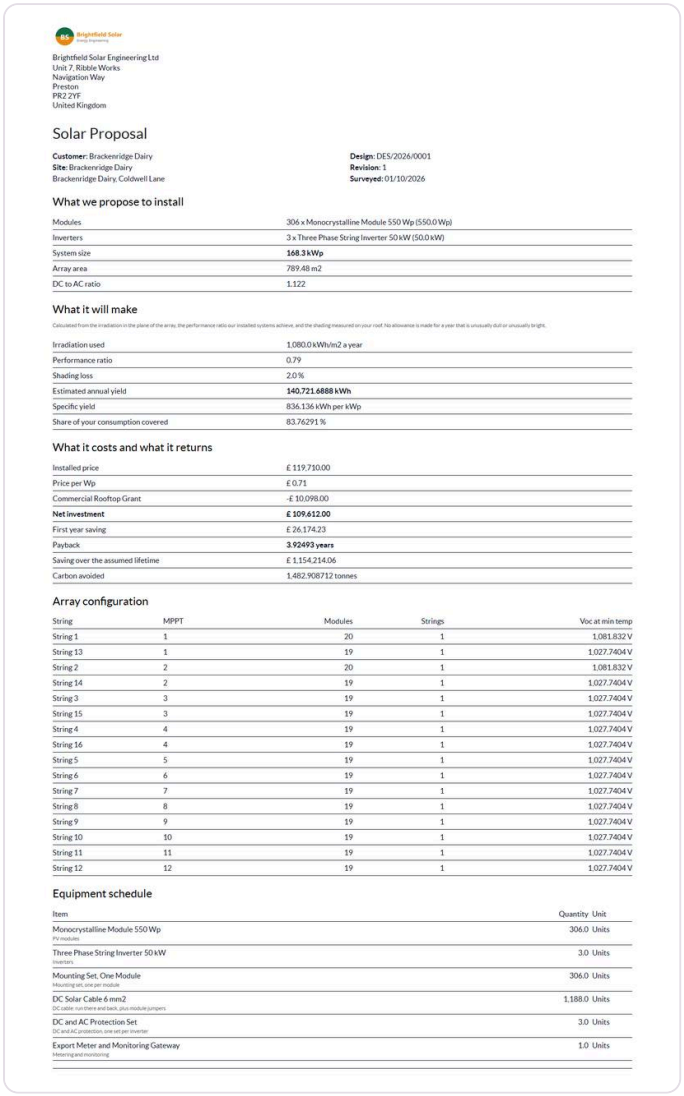
The project pivot

The board as numbers: kWp and value by stage or state.

> Costs are entered on the job; they are not pulled from vendor bills or timesheets. Milestone invoices are real Odoo invoices, so receivables and payments live in Accounting as usual.

PART 15 Printed documents, emails and the nightly jobs

Four documents print from the records, three emails can go to the customer, and five scheduled jobs keep the dates honest.



Brightfield Solar Engineering Ltd
Unit 7, Ribble Works
Navigation Way
Preston
PR2 2YF
United Kingdom

Solar Proposal

Customer: Brackenridge Dairy
Site: Brackenridge Dairy
Brackenridge Dairy, Coldwell Lane

Design: DES/2026/0001
Revision: 1
Surveyed: 01/10/2026

What we propose to install

Modules	306 x Monocrystalline Module 550 Wp (550.0 Wp)
Inverters	3 x Three Phase String Inverter 50 kW (50.0 kW)
System size	**168.3 kWp**
Array area	789.48 m2
DC to AC ratio	1.122

What it will make

Calculated from the irradiation in the plane of the array, the performance ratio our installed systems achieve, and the shading measured on your roof. No allowance is made for a year that is unusually dull or unusually bright.

Irradiation used	1,080.0 kWh/m2 a year
Performance ratio	0.79
Shading loss	2.0 %
Estimated annual yield	**140,721.6888 kWh**
Specific yield	836.136 kWh per kWp
Share of your consumption covered	83.76291 %

What it costs and what it returns

Installed price	£ 119,710.00
Price per Wp	£ 0.71
Commercial Rooftop Grant	-£ 10,098.00
Net investment	**£ 109,612.00**
First year saving	£ 26,174.23
Payback	**3.92493 years**
Saving over the assumed lifetime	£ 1,154,214.06
Carbon avoided	1,482.908712 tonnes

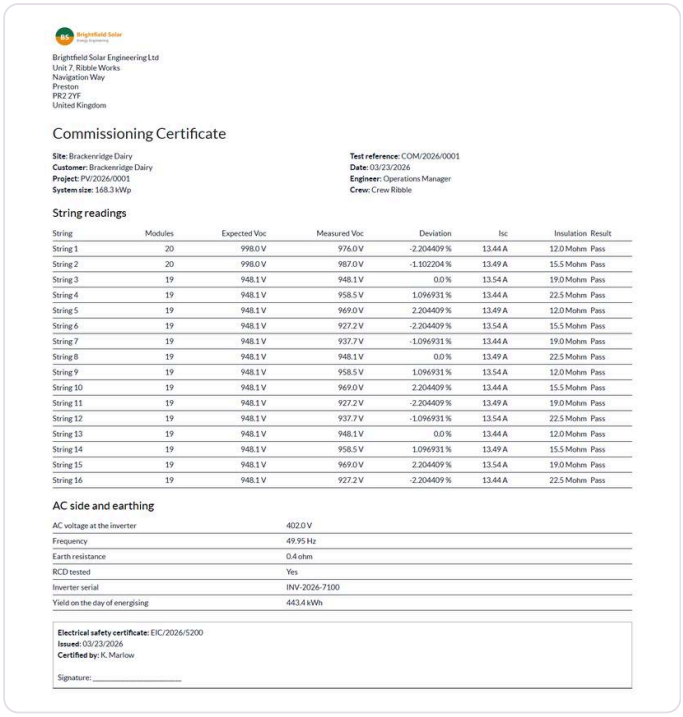
Array configuration

String	MPPT	Modules	Strings	Voc at min temp
String 1	1	20	1	1,081.832 V
String 13	1	19	1	1,027.7404 V
String 2	2	20	1	1,081.832 V
String 14	2	19	1	1,027.7404 V
String 3	3	19	1	1,027.7404 V
String 15	3	19	1	1,027.7404 V
String 4	4	19	1	1,027.7404 V
String 16	4	19	1	1,027.7404 V
String 5	5	19	1	1,027.7404 V
String 6	6	19	1	1,027.7404 V
String 7	7	19	1	1,027.7404 V
String 8	8	19	1	1,027.7404 V
String 9	9	19	1	1,027.7404 V
String 10	10	19	1	1,027.7404 V
String 11	11	19	1	1,027.7404 V
String 12	12	19	1	1,027.7404 V

Equipment schedule

Item	Quantity	Unit
Monocrystalline Module 550 Wp PV modules	306.0	Units
Three Phase String Inverter 50 kW Inverters	3.0	Units
Mounting Set, One Module Mounting set, one per module	306.0	Units
DC Solar Cable 6 mm2 DC cable, run there and back, plus module jumpers	1,188.0	Units
DC and AC Protection Set DC and AC protection, one set per inverter	3.0	Units
Export Meter and Monitoring Gateway Metering and monitoring	1.0	Units

The system proposal, from the design.



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United Kingdom

Commissioning Certificate

Site: Brackenridge Dairy
Customer: Brackenridge Dairy
Project: PV/2026/0001
System size: 168.3 kWp

Test reference: COM/2026/0001
Date: 03/23/2026
Engineer: Operations Manager
Crew: Crew Ribble

String readings

String	Modules	Expected Voc	Measured Voc	Deviation	Isc	Insulation	Result
String 1	20	998.0 V	976.0 V	-2.204409 %	13.44 A	12.0 Mohm	Pass
String 2	20	998.0 V	987.0 V	-1.102204 %	13.49 A	15.5 Mohm	Pass
String 3	19	948.1 V	948.1 V	0.0 %	13.54 A	19.0 Mohm	Pass
String 4	19	948.1 V	958.5 V	1.096931 %	13.44 A	22.5 Mohm	Pass
String 5	19	948.1 V	969.0 V	2.204409 %	13.49 A	12.0 Mohm	Pass
String 6	19	948.1 V	927.2 V	-2.204409 %	13.54 A	15.5 Mohm	Pass
String 7	19	948.1 V	937.7 V	-1.096931 %	13.44 A	19.0 Mohm	Pass
String 8	19	948.1 V	948.1 V	0.0 %	13.49 A	22.5 Mohm	Pass
String 9	19	948.1 V	958.5 V	1.096931 %	13.54 A	12.0 Mohm	Pass
String 10	19	948.1 V	969.0 V	2.204409 %	13.44 A	15.5 Mohm	Pass
String 11	19	948.1 V	927.2 V	-2.204409 %	13.49 A	19.0 Mohm	Pass
String 12	19	948.1 V	937.7 V	-1.096931 %	13.54 A	22.5 Mohm	Pass
String 13	19	948.1 V	948.1 V	0.0 %	13.44 A	12.0 Mohm	Pass
String 14	19	948.1 V	958.5 V	1.096931 %	13.49 A	15.5 Mohm	Pass
String 15	19	948.1 V	969.0 V	2.204409 %	13.54 A	19.0 Mohm	Pass
String 16	19	948.1 V	927.2 V	-2.204409 %	13.44 A	22.5 Mohm	Pass

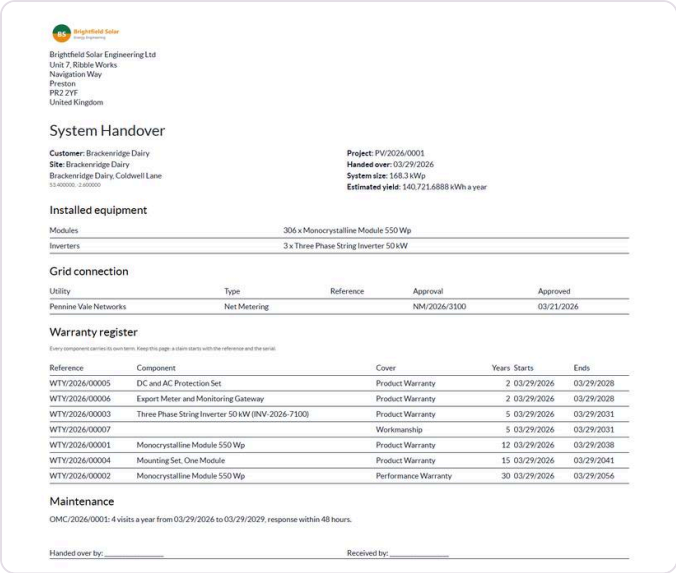
AC side and earthing

AC voltage at the inverter	402.0 V
Frequency	49.95 Hz
Earth resistance	0.4 ohm
RCD tested	Yes
Inverter serial	INV-2026-7100
Yield on the day of energising	443.4 kWh

Electrical safety certificate: EIC/2026/5200
Issued: 03/23/2026
Certified by: K. Marlow

Signature: _____

The commissioning certificate.



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Preston
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United Kingdom

System Handover

Customer: Brackenridge Dairy
Site: Brackenridge Dairy
Brackenridge Dairy, Coldwell Lane
53.400000, -2.600000

Project: PV/2026/0001
Handed over: 03/29/2026
System size: 168.3 kWp
Estimated yield: 140,721.6888 kWh a year

Installed equipment

Modules	306 x Monocrystalline Module 550 Wp
Inverters	3 x Three Phase String Inverter 50 kW

Grid connection

Utility	Type	Reference	Approval	Approved
Pennine Vale Networks	Net Metering		NM/2026/3100	03/21/2026

Warranty register

Every component carries its own term. Keep this page: a claim starts with the reference and the serial.

Reference	Component	Cover	Years	Starts	Ends
WTY/2026/00005	DC and AC Protection Set	Product Warranty	2	03/29/2026	03/29/2028
WTY/2026/00006	Export Meter and Monitoring Gateway	Product Warranty	2	03/29/2026	03/29/2028
WTY/2026/00003	Three Phase String Inverter 50 kW (INV-2026-7100)	Product Warranty	5	03/29/2026	03/29/2031
WTY/2026/00007		Workmanship	5	03/29/2026	03/29/2031
WTY/2026/00001	Monocrystalline Module 550 Wp	Product Warranty	12	03/29/2026	03/29/2038
WTY/2026/00004	Mounting Set, One Module	Product Warranty	15	03/29/2026	03/29/2041
WTY/2026/00002	Monocrystalline Module 550 Wp	Performance Warranty	30	03/29/2026	03/29/2056

Maintenance

OMC/2026/0001: 4 visits a year from 03/29/2026 to 03/29/2029, response within 48 hours.

Handed over by: _____ Received by: _____

The handover pack.



Brightfield Solar Engineering Ltd
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Preston
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United Kingdom

Daily Site Report

Report: DSR/2026/00003
Site: Brackenridge Dairy
Project: PV/2026/0001

Date: 02/26/2026
Crew: Crew Ribble
Weather: Clear

On site	4 people for 8.0 hours (32.0 labour hours)
Modules installed today	51

Work done
Rails set out and modules fixed on the centre bank. DC strings landed in the combiner and labelled.

Safety incident
Near miss: a rail slid on the sheet while being passed up. Toolbox talk held, edge protection extended.

Signed: Operations Manager _____

A daily site report.

EMAIL	SENT BY
Solar: Proposal Sent	Send Proposal on a job
Solar: Maintenance Visit Due	the visits coming up job
Solar: Site Underperforming	the below estimate job

JOB	RUNS	DOES	SHIPS
Solar: milestones due and paid	daily	Moves milestones to Due and Paid, refreshes the late flag	on
Solar: permits and certificates about to lapse	weekly	A to do on documents expiring soon	on
Solar: warranty cover running out	weekly	A to do on warranties expiring soon	on
Solar: maintenance visits coming up	daily	Reminders and emails for visits	off
Solar: sites generating below estimate	weekly	Raises underperforming sites	off

The two jobs that send mail ship switched off; switch them on in Settings, Technical, Scheduled Actions (developer mode) once outgoing mail is set up. The demo has no outgoing mail server.

PART 16 Who sees what, a walk-through and questions

The access rules in one table, then the order to show the system to somebody else in, and the questions people ask.

RULE	EFFECT
A technician sees their crew	Site Technicians see the jobs, daily reports, maintenance visits and commissioning tests of the crews they belong to
Money is for engineers	Contract value, costs, prices, invoicing and the payment schedule are hidden from technicians; margins and project costs are for managers
Meter numbers	The meter number on a grid application is for engineers and managers
Configuration and reporting	Solar Manager only
Companies	Every record belongs to a company and each company sees only its own

A fifteen minute walk-through

#	DO THIS	AND NOTICE
1	Delivery, Projects board	*Ten jobs across the stages; a red Late tag on Elmcroft Care Home.*
2	Drag the Redbrook card from Survey to Installation	*Refused: the stage gate wants the documents of every stage it would skip.*
3	Delivery, Site Surveys, the first survey	*Usable area, the size it fits, the bills and the structural sign off.*
4	Delivery, System Designs, DES/2026/0001, Array	*Strings sized against the inverter at the coldest morning.*
5	Its Yield and Savings tab	*Yield, bill offset, payback walked year by year, carbon avoided.*
6	Its Parts List tab	*Modules, inverters, mounting, cable, protection and meter, priced.*
7	PV/2026/0001, Payment Schedule	*Four milestones adding up to 100 percent, invoiced and paid.*
8	Compliance, Grid Applications	*One refused for transformer headroom, resubmitted at 30 kW.*
9	Compliance, Commissioning	*A passed test with every string read, and a failed one with the reason.*
10	Compliance, Warranty Register	*Product, performance and workmanship cover per component, one claim open.*
11	O&M Contracts, then Underperforming Sites	*The visit calendar, and a site making 82 percent of estimate.*
12	Reporting, Margin by Project	*Contract value against estimate and actual cost per job.*

Questions people ask

QUESTION	ANSWER
Does it need Odoo Enterprise?	No. It runs on Odoo 19 Community with Inventory and Invoicing, which are part of Community.

Does it draw the array or model shading?	No. It takes the area, obstructions and shading loss from the survey. Layout drawings and shading studies are filed as documents.
Does it read inverters or monitoring portals?	No. Meter readings are entered or imported into the generation log.
Does it file applications with the utility?	No. It records the application, the dates, the fee and the approval; lodging it is still done on the utility's own portal.
Can customers log in?	No. There is no customer portal in the module.
Does it handle batteries?	A product can be marked as battery storage and go on the parts list by hand, but the design does not size batteries.
Can one database run several companies?	Yes. Every record carries a company and each company sees only its own.

PART 17

The menu map

Every screen in the module, on one page.

Delivery

Projects	Board by stage, commercial state, money and dates
Site Surveys	Roof, bills, structure
System Designs	Strings, yield, parts list, price, payback
Daily Site Reports	Crew, weather, modules up, incidents

Compliance

Grid Applications	Lodged, inspected, approved or refused
Project Documents	The papers each stage gate asks for
Commissioning	String readings and the safety certificate
Warranty Register	Per component, with claims

Operations and Maintenance

O&M Contracts	Fee, visits, response, guaranteed PR
Maintenance Visits	Scheduled, completed, missed
Generation Log	Metered against expected
Underperforming Sites	Below the threshold, not reviewed

Billing

Payment Schedule	Milestones due, invoiced, paid
Project Costs	What each job actually cost

Reporting

Margin by Project	Value, estimate, actual, margin
Maintenance Revenue	Fees and kWp under contract

Configuration

Settings	Yield defaults, gates, tolerances, warnings
Project Stages	Nine stages and their documents
Utilities	Export cap, fee, lead days, inspection
Incentive Schemes	Per kWp, percent or fixed
Installation Crews	Capacity and accreditation
Solar Components	Modules, inverters, balance of system

Want this for your solar business?

We set it up from your component list, your stages and your utility's rules, load your live jobs and your sites under contract, connect milestone invoicing to your accounts, and train the design engineers, project managers and crews.

way4tech.com · info@way4tech.com